NEAREST-NEIGHBOUR METHODS FOR TIME SERIES ANALYSIS

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Abstract. The nearest-neighbour method, because of its intuitively appealing nature and competitive theoretical properties, deserves consideration in time-series applications akin to attention it has received lately in the i.i.d. case. Here it is shown that as a nonparametric regression device, like the kernel method, under the G_2 mixing assumption, it converges in quadratic mean at the Stone-optimal rate. In the closing sections, our methodology is extended to a broader pattern-recognition context, and applied to hydrologic data.

Keywords. Time series; nonparametric estimation; nearest-neighbour methods; nonlinear prediction.

1. INTRODUCTION

The object of study is a stationary ergodic time series $\{X(n)\}$ which, in the next two sections, will be further assumed Markov. It is presumed that the time series has a stationary pdf $\pi(x)$ for the unconditional distribution of $X_i(n)$, and a transition pdf $f(y \mid x)$ for the conditioned variable $X(n+1) \mid (X(n) = x)$. The statistical objective will be to infer the non-linear regression function

$$m(x) = E[X(n+1) \mid X(n) = x] \tag{1.1}$$

from an observation of an initial portion of the series.

The method to be analysed is the $k(n)$-nearest-neighbour (k-NN) method which we now define. Let $\{k(n)\}$, $n > 1$, be an unbounded nondecreasing sequence of positive integers with $k(n) < n$, and for each n, take $\bar{R}(n)$ to be the radius of the smallest ball centred at state x which contains $k(n)$ of the states $X(j)$, $j < n$. For any $r > 0$, let $B(r)$ be the ball of radius r centred at x. Take RO to be some relatively large number, and $R(n) \equiv$ minimum $\{\bar{R}(n), RO\}$. We adopt the following version of the NN rule.

DEFINITION. Let $B_n = B(R(n))$ be the ball of radius $R(n)$ centred at x. The k-NN estimate of $m(x)$ is given by

$$m_n(x) = 1/(k(n)) \sum X(j+1) : j \in S(n) \tag{1.2}$$

where $S(n) = \{j : \mid X(j) - x \mid \leqslant R(n)\}$.

Here we presume that j ranges from 1 to $n-1$, and so $m_n(x)$ is determined entirely by the initial n-segment of $\{X(i)\}$.

The artifice of bounding the distance to the furthest neighbour will simplify section 2, but as is seen below, almost surely $\bar{R}(n) > RO$ for at most finitely many

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n. Except for this artifice, our definition coincides with that of earlier pattern-recognition papers, but Stone (1977) and Devroye (1978, 1982) have proposed more inclusive versions.

Let RO be any positive number as in the definition (1.2). Amost surely, there is an integer $N = N(RO)$ such that for all $n > N$, $\bar{R}(N) < RO$. For under our assumptions, the process is ergodic and so (Doob, 1948) if $N(RO, n)$ denotes the number of observations $X(i)$, $1 \leqslant i < n$, which lie within the ball $B(RO)$, then with probability 1,

$$N(RO, n)/n \to \pi(B(RO)). \tag{1.3}$$

In (1.3) and elsewhere, if E is a set, $\pi(E)$ is its π probability. Since $k(n)/n \to 0$, eventually (say by epoch N), $k(n) < N(RO, n)$ for all $n > N$, provided only that $\pi(B(RO)) > 0$.

In section 2 on convergence of $m_n(x)$ in the mean square, we will further assume that $\pi(x)$, $m(x)$, and $f(y \mid x)$ are twice continuously differentiable and $f(y \mid x)$ is bounded. Moreover, in section 2, it will be presumed that $\{X(i)\}$ satisfies the Rosenblatt (1970) G_2 mixing condition that for some integer v and number ρ in the open unit interval, and for any bounded continuous function $h(x(1), x(2), \ldots)$, with π-expectation 0,

$$E[E[h(X(v), X(v+1), \ldots) \mid X(1)]^2] \leqslant \rho E[h(X(1), X(2) \ldots)^2]. \tag{1.4}$$

For convenience, we will assume v, above, equals 2. Finally, we impose the condition that $\operatorname{var}(X(N+1) \mid X(n) = x)$ exists and is smooth.

The purpose of section 2 is to show that under these conditions, $m_n(x)$ converges in the quadratic mean at the Stone optimal rate. The complements analogous results for the kernel non-parametric regression method under these Markov sequence assumptions obtained by Robinson (1983) and, even closer, by Yakowitz (1985a). In section 3, results of the preceding section are extended beyond Markovian to more general time series settings. The plan is to set the stage for bringing pattern-recognition notions such as 'feature vectors' to bear on time series decision problems. Section 4 gives an application of NN methodology to analysing hydrologic data.

Our developments will be seen to be inspired by results and techniques of Mack and Rosenblatt (1979), Mack (1980, 1981), and Rosenblatt (1970). Other recent incisive studies of the nearest-neighbour method include Devroye (1978, 1981, 1982). These works are concerned with strong and L_p convergence. All studies above are concerned with k-NN estimators of $m(x) = E[y \mid X = x]$, based on i.i.d pairs $\{(X_i, Y_i)\}_{i=1}^n$. Collomb (1984a, b) has studied k-NN estimates in the context of time series, showing uniform pointwise convergence under a ϕ-mixing assumption.

Our practical motivation here, following Yakowitz *et al.* (1983), is to provide foundation for applying nearest-neighbour methods to decision problems in stochastic hydrology. A riverflow application of the kernel method is described in Yakowitz (1985b), and Yakowitz *et al.* (1983) gives initial NN studies of riverflow decision problems. This latter work announces the results proven in the present

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study. In his dissertation, Karlsson (1985) has applied nearest-neighbour regression to rainfall/runoff analysis. The closing section of the present work outlines one of his studies.

2. CONVERGENCE IN QUADRATIC MEAN

The purpose of this section is to show that under the hypotheses of G_2 mixing and second-order differentiability, the NN non-parametric regression estimator is convergent in quadratic mean, and provided $\{k(n)\}$ is defined according to specifications below, achieves the Stone (1980) optimal rate for the order of differentiability we have assumed. The state space $\mathscr{X}$ for our Markov sequence is presumed to be a subset of R^d. Define $\tilde{Q} = 4/(4 + d)$ and take

$$k(n) = \text{integer part of } n^{\tilde{Q}}. \tag{2.1}$$

On the left side of (2.2) below and elsewhere, we will freely use the notation of real (as opposed to vector-valued) functions and variables, hoping that the corresponding expressions for multivariate terms will be obvious to the reader from the context.

THEOREM 2.1. *Assume $\{X(i)\}$ is G_2 ergodic Markov, and $\pi(x)$, $m(x)$, $f(y \mid x)$ and $v(x) = \text{var}(X(n+1) \mid X(n) = x)$ are twice continuously differentiable. If $f(y \mid x)$ is bounded, then for x in the support of π, the expected squared estimation error almost surely satisfies*

$$E[(m_n(x) - m(x))^2] = O(n^{-\tilde{Q}}). \tag{2.2}$$

Below and throughout, we will continue to use the notation $B(r)$ to denote the ball of radius r centred at x. Assuming $k(n)$ is specified, as in section 1, $R(n)$ denotes the smaller of RO and the distance from x to its kth nearest neighbour. The following statement about the second central moment of the radii $R(n)$ is a key to proving theorem 2.1.

LEMMA. *Under the preceding assumptions,*

$$E[R(n)^2] = O(n^{(-\tilde{Q}/2)}). \tag{2.3}$$

PROOF OF THE LEMMA. For each n, we define a radius $r(n)$ so that for $N(r, n)$ the number of state observations $X(i)$, $i < n$, such that $|X(i) - x| \leqslant r$, we have

$$E[N(r(n), n)] = 2k(n) = 2 \text{ int}(n^{\tilde{Q}}), \tag{2.4}$$

with $k(n)$ as in (2.1). By using a result of Rosenblatt (1970), it will transpire that

$$r(n) = O(n^{(-\tilde{Q}/4)}) \quad \text{and} \quad \text{var}(N(r(n), n)) = O(n^{\tilde{Q}}). \tag{2.5}$$

Assuming the validity of (2.5), from the Chebyshev inequality we have

$$\begin{aligned} P[R(n) > r(n)] &= P[2k(n) - N(r(n), n)) > k(n)] \leqslant \text{var}(N(r(n), n))/k(n)^2 \\ &= O(n^{\tilde{Q}})/n^{2\tilde{Q}}. \end{aligned} \tag{2.6}$$

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So $P[R(n) > r(n)] = O(n^{-Q})$. Now, since $R(n)$ is constrained to be less than RO

$$E[R(n)^2] \leqslant r(n)^2 + RO^2 P[R(n) > r(n)] = O(n^{(-Q/2)}) + O(n^{-Q}). \tag{2.7}$$

Thus the proof rests on showing (2.5), to which our attention is now devoted. □

Let $w(x)$ be a bounded continuous pdf. Then from Rosenblatt (1970), and as placed into the multivariate setting by Yakowitz (1985a), under our assumptions about the $\{X(n)\}$ process, the kernel non-parametric density estimator

$$f_n(x) = (1/(nr^d)) \sum_{i=1}^{n} w((X(i) - x)/r) \tag{2.8}$$

satisfies

$$E[f_n(x)] - \pi(x) = O(r^2) \tag{2.9}$$

and

$$\operatorname{var}(f_n(x)) = O((nr^d)^{-1}). \tag{2.10}$$

Suppose that $w(x)$ is chosen to be continuous, to be bounded by 1, and for some $\varepsilon > 0$, satisfies

$$w(x) \begin{cases} 0, & |x| > 1 + \varepsilon, \\ 1/c(d), & |x| < 1 - \varepsilon, \end{cases}$$

$c(d)$ being the volume of the unit sphere.

Then $w(x)$ satisfies the hypotheses on which (2.8) to (2.10) depend. Thus we may conclude that if $1_\varepsilon(x)$ is the indicator function for $||x| - 1| < \varepsilon$, then

$$\left| f_n(x) - \frac{N(r, n)}{c(d)nr^d} \right| \leqslant 2 \frac{\sum 1_\varepsilon((X_i - x)/r)}{c(d)nr^d}. \tag{2.11}$$

Upon taking expectations and using (2.9), we have,

$$|E[N(r, n)] - \pi(x)c(d)nr^d| \leqslant [O(r^2) + O(\varepsilon)]c(d)nr^d. \tag{2.12}$$

Since ε can be made arbitrarily small, the second summand may be regarded as negligible. Upon choosing r to satisfy (2.4), with $k(n)$ as in (2.1), (2.12) yields

$$r(n) = O(n^{(-1/(d+4))}). \tag{2.13}$$

This demonstrates the first part of the key equation (2.5). The second part follows in the same fashion from (2.10) and (2.11), and we consider the proof of the lemma to be complete. In view of the fact (shown in section 1) that $\bar{R}(n) > RO$ only finitely often and the 'almost surely' qualification of the theorem, we will justifiably ignore the 'RO' construct in the remainder of this study.

PROOF OF THE THEOREM. Our next objective is to use the lemma to prove that the bias and variance of $m_n(x)$ converge as required by the theorem. Toward that end, we examine explicit formulae for densities and expectations.

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For each i, $1 \leqslant i \leqslant k$, let $N(i)$ denote the index of the ith (in time) among the k-NNs. For now, fix i and define $\bar{X} = X_{N(i)}$, $Y = Y_{N(i)+1}$. Assume that Y is not the furthest of the k-NNs. That is, $|Y| < R(n)$. The joint pdf $p(\bar{x}, y)$ can be factored as

$$p(\bar{x}, y) = p(\bar{x}) f(y \mid \bar{x}). \tag{2.14}$$

Then for $\tilde{m} = E[Y]$, we have that

$$\begin{aligned} m(x) - \tilde{m} &= \iint y(f(y \mid x) - f(y \mid \bar{x}))p(\bar{x})\, dy\, d\bar{x} \\ &= \int (m(x) - m(\bar{x}))p(\bar{x})\, dx. \end{aligned} \tag{2.15}$$

Now observe that

$$p(\bar{x} \mid \bar{x} \in B(r)) = \pi(\bar{x})/\pi(B(r)),$$

and use the differentiability of $\pi(\cdot)$ and $m(\cdot)$ to conclude that

$$m(x) - E[\tilde{m} \mid \bar{X} \leqslant r] = O(r^2). \tag{2.16}$$

This order obtains for any bounded random variable $\bar{X}$ with once differentiable pdf, and twice-differentiable integrand $m(\cdot)$; a demonstration involves examination of the representation

$$\begin{aligned} \frac{1}{\pi(B(r))} \int_{B(r)} \{[m'(x)(x - \bar{x}) + \tfrac{1}{2}m''(x)(x - \bar{x})^2] \\ \times [\pi(x) + \pi'(x)(\bar{x} - x)] + o(|x - \bar{x}|^2)\}\, d\bar{x}, \end{aligned} \tag{2.17}$$

noting that the linear term vanishes because $B(r)$ is symmetric.

The order (2.16), the NN definition constraint of section 1 that $R \leqslant RO$, and the lemma allow us to conclude that if $X_{N(i)}$ is not followed by the furthest nearest neighbour (NN), then the bias is $O(n^{-Q/2})$. The probability of this last contingency must decrease as $1/k = O(n^{-Q})$, since at most one NN can be followed by the furthest NN. In summary, we have shown

$$E[m_n(x)] - m(x) = O(n^{(-Q/2)}). \tag{2.18}$$

The mean squared error $E[(m_n(x) - m(x))^2]$ of course can be decomposed as

$$E[(m_n(x) - m(x))^2] = \operatorname{var}(m_n(x)) + (\text{bias})^2. \tag{2.19}$$

Having already considered the bias term, our attention now passes to the variance. Using results and notation from our examination of the bias, we can conclude that,

$$\operatorname{var}(Y) = v(x) + O(n^{-Q/2}). \tag{2.20}$$

The theorem will follow if only we can show that the total influence of covariance terms is asymptotically negligible. As above, let $N(1)$ denote the index of the first

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(in time) nearest-neighbour, $N(2)$ the second, etc. Our intention is to show that

$$\sum_{v>i} |\operatorname{cov}(X(N(v)+1), X(N(i)+1))| \to O \qquad \text{as} \quad n \to \infty \quad \text{and} \quad r \to 0. \tag{2.21}$$

From an argument given by Rosenblatt (1970), the G_2 assumption implies that in terms of (1.4),

$$|\operatorname{cov}(X(N(i)+1+v), X(N(i)+1))|^2 \leqslant \rho^v \operatorname{var}(X(N(i))). \tag{2.22}$$

Equation (2.22) implies (2.21) if only we can show that

$$P[N(i+1) - N(i) < v] \to 0 \tag{2.23}$$

as $n \to \infty$. We demonstrate this for $v = 1$, and the reader will see how our argument may be extended to an arbitrary integer v. Using $\bar{X}$, and Y as defined in connection with our analysis of bias, and letting $P_{\bar{x}, y}(R > |y - x|)$ be the conditional (on specified $\bar{x}$ and y outcomes) that the distance to the furthest NN is greater than the distance to the successor of $X(N(i))$, we have

$$P[N(i+1) = N(i) + 1] = \int P_{\bar{x}y}[R > |y - x|] f(y|\bar{x}) p(\bar{x})\, d\bar{x}\, dy. \tag{2.24}$$

But by hypothesis, $f(\cdot)$ is bounded, and from the lemma, for any $a > 0$,

$$P_{\bar{x}, y}[R > a] \to 0 \qquad \text{as} \quad n \to \infty$$

for any $(\bar{x}, y)$ in the support of $\pi(\bar{x}) f(y|\bar{x})$. These observations yield (2.23), for $v = 1$.

These two facts, (2.22) and (2.23) and the Lebesgue convergence theorem jointly imply (2.21), which, in turn, implies that, almost surely

$$\operatorname{var}(m_n(x)) = O(1/k) = O(n^{-\bar{Q}}). \tag{2.25}$$

In summary, from (2.18), (2.19), and (2.21), $E[m_n(x) - m(x))^2] = O(n^{-\bar{Q}})$, and the theorem is proven.

We note that the asymptotic negligibility of the covariance terms implies that the variance of $(k(n))^{1/2}(m_n(x) - m(x))$ converges to that of the i.i.d. case, which has been shown by Mack (1981) to be

$$v(x) = \operatorname{var}(X(n+1)|X(n) = x),$$

in our notation. The asymptotic bias expression of that paper need not hold under our assumptions, since (in contrast to the i.i.d. case) the constant in the order of convergence in (2.3) is not known and is in all likelihood process-dependent.

Let us tie the preceding to results by Stone (1980) on optimal rates of convergence. We transpose that study to our notation. Let $\{(X(i), Y(i))\}$ denote an i.i.d. sequence of random d-dimensional pairs. M is a collection of 'regression' functions $m(x)$. Stone defined a rate r to be 'achievable' with respect to M, if, for

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some sequence $\{m_n(x)\}$, with $m_n(x)$ constructed entirely from an (X, Y) sample of size n,

$$\lim_{C \uparrow \infty} \lim \sup_n \sup_m P_m(|m_n(x) - m(x)| > Cn^{-r}) = 0.$$

In this paper, we are not dealing with an i.i.d. sequence but one can still interpret our type of regression problem within Stone's framework by considering the random pairs $\{(X(i), X(i+1))\}$. Successive pairs are, of course, no longer i.i.d. Stone's theorem 1 implies that if M is the class of functions differing from some fixed twice-differentiable function by some second-order Lipschitz function, with Lipschitz constant uniform over all x and all $m(x) \in M$, and if $Y(X)$ has a second moment, then the highest achievable rate is $\tilde{Q}/2$. In view of our theorem 2.1, the Chebyshev inequality implies that we are achieving this optimal rate. Stone's class M is larger, but a review of our derivation of bias convergence suggests that our result could likewise be enlarged to that M. Our result is essentially weaker in that Stone does not require the existence, much less the differentiability, of $f(y|x)$, nor the existence of a pdf for X. Yet, to our knowledge, nobody has proven that strengthening the assumptions to include these smoothness conditions yields a faster rate in the i.i.d. case. In this sense, we suspect that the NN regression estimator cannot be improved for G_2-mixed time series and twice-differentiable regression functions. (The same goes for the kernel time series estimate (Yakowitz, 1985a).)

3. BEYOND THE MARKOV ASSUMPTION TOWARD FEATURE VECTORS

Practitioners tend to apply ARMA methodology to data sets of interest without concerning themselves about tests to see whether the data are consistent with the somewhat stringent ARMA hypotheses. Indeed there does not seem to be much theoretical foundation available for such tests, anyhow. (Practitioners usually perform tests for the proper AR and MA orders, but this is another matter.) While this author is skeptical about such ARMA modelling, he has a certain admiration for the audacity and pragmatism of this approach. (The scepticism arises from his awareness of easily-contrived examples in which ARMA decisions are not asymptotically optimal.) In his studies of geostatistical and stock market data sets, the author has applied NPR techniques with similar abandon. But this cavalier onslaught was fuelled by the notion that non-parametric regression methods are already known to yield asymptotically optimal decision functions under relatively lenient conditions, and in the absence of solid counterexamples showing inconsistency, there is reason to hope that theoretical justifications eventually will be found for what now may appear as indiscriminant practice. We mention, however, that extension of ARMA ideas to nonlinear models, as in Haggan *et al.* (1984), offers considerable promise toward enlarging the parametric framework. In fact, these broader parametric models, when incorporated into a hierarchical structure with a testing procedure in the spirit of Shibata (1984), appear to this writer as a promising avenue of inquiry.

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The aim of some of our present research is to push ahead toward establishing more lenient convergence conditions that do not require the Markov assumption or even stationarity, and in fact, require little more than asymptotic independence. The intuitive content is that we would like to justify the pattern recognition practice of basing decision rules on whatever low-dimension 'feature vector' statistic seems maximally 'informative' with respect to the time-series decision problem at hand.

Collomb (1984a, b), Rosenblatt (1970), Robinson (1983) and Yakowitz (1985a) have already given some consideration to general time series, the latter paper giving conditions which include ARMA processes with bounded independent white noise sequences. The discussion we now give regarding quadratic convergence, however, is inspired by the 'S' condition offered in the closing portion of Rosenblatt's (1970) study. Here, Rosenblatt has observed that the kernel density estimate of the marginal pdf $\pi(x)$ converges as in the i.i.d. case whenever $\{X(i)\}$ is stationary and satisfies the mixing condition that for some number N and positive sequence $\{a(k)\}$ which is $O(k^{-q})$, and $q > 4$,

$$E[E(g(X(N+k), X(N+k+1), \ldots) \mid X(j), 1 < j < N]^2 < a(k)\text{var}(g) \qquad (3.1)$$

for any random variable g with mean 0 and finite variance. In light of Rosenblatt's observation that under condition S, the non-parametric density estimator converges as in the G_2 case, we may conclude for the same reasons as before that (2.5) is valid, and this in turn led to our section 2 conclusion that the squared bias is $O(n^{-\tilde{Q}})$. In the non-Markov case, we must consider pdfs $f(y \mid x)$ for the conditional variable $X(n+1) \mid (\boldsymbol{X}(n) = \boldsymbol{x})$ which depend on the entire past. Thus $\mathbf{x}$ gives the observed trajectory up to time n, and $\boldsymbol{X}(n) = \{X(j), j \leqslant n\}$. The convariance terms become asymptotically negligible if we assume, in the spirit of section 2, that the conditional pdf $f(y \mid \boldsymbol{x})$ is bounded, for the same reasons as we gave earlier in connection with (2.19). Thus we have shown the following:

COROLLARY (To theorem 2.1). If $\{X(i)\}$ is stationary and satisfies (3.1) for some number N and sequence $\{a(k)\}$ for which $\sum a(k) < \infty$, and if the marginal pdf $\pi(x)$ for $X(i)$ and the conditional pdf $f(\cdot \mid \cdot)$ for $X(n+1) \mid (\boldsymbol{X}(n) = \boldsymbol{x})$ are twice continuously differentiable, and if the variance associated with $f(y \mid \boldsymbol{x})$ is continuous in $\mathbf{x}$, then for $\tilde{Q} = 4/(d+4)$ and $\pi(x) > 0$,

$$E[(m_n(x) - m(x))^2] = O(n^{-\tilde{Q}}),$$

$m_n(x)$ being given by the NN rule (1.2).

4. APPLICATION OF NEAREST-NEIGHBOUR METHODS TO RUNOFF PREDICTION

This section relates a computational experiment undertaken by Karlsson (1985). Related studies are described in Yakowitz and Karlsson (1985, 1986).

The data set for our computational investigation was rainfall/runoff measurements collected by the National Weather Service from the Bird Creek Ohio watershed, which drains an estimated 2344 km area. Our record consists of 2400

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runoff and 9600 rainfall observations, the runoff having been recorded at daily intervals and the rainfall, at six-hour frequency. The data was divided into split samples, the first 2000 runoff points being used for model calibration, and the remaining 400 points for assessment of performance. In Figure 1, we give graphs of portions of the rainfall and runoff calibration data.

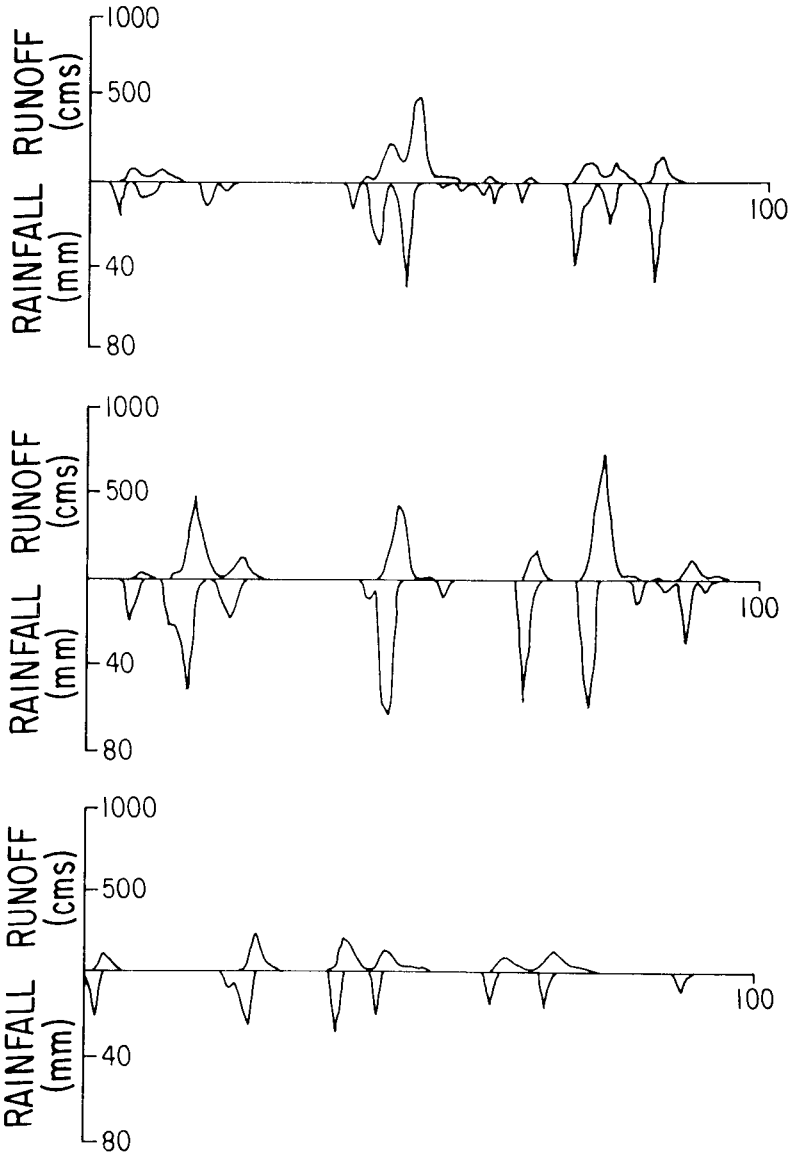


FIGURE 1. Three month season from three successive years of Bird Creek Data.

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Our objective was to compare NN performance with standard procedures. The 'standard' procedures here are the ARMAX model (e.g., Ljung and Soderstrom 1983), which is a standard linear system model, and the 'Sacramento model', a physically based rainfall–runoff model constructed by Burnash and Ferral (1980) of the National Weather Service. It is a tricky model to understand and calibrate, but fortunately the National Weather Service, in collaboration with Dr. Sorooshian and colleagues at the University of Arizona, has already produced a calibration for the Bird Creek data—in fact, this was our reason for selecting that data set.

Toward calibrating the NN method, the 2000 point learning set was further split into subsets of 1600 and 400 runoff values, respectively. Somewhat arbitrarily, a feature vector consisting of the past three days of runoff and past two rainfall observations was selected. We denote the feature vector consisting of 3 rainfall and two runoff observations of day n by

$$Z(n) = Y((n),\ Y(n-1),\ Y(n-2),\ X(n),\ X(n-1)). \tag{4.1}$$

For any 5-tuple of positive weights $w = (w(1), \ldots, w(5))$, one defines a norm on feature vectors by

$$\| z \|_w^2 = \sum_{i=1}^{5} w(i)^* z(i)^2. \tag{4.2}$$

The norm quantifies 'nearness' of vectors. By using the first 1600 observations to construct the NN-predictors

$$\tilde{m}(z;\ w,\ k)$$

for any weight vector w and nearest-neighbour number k one may assign a performance measure using the remaining 400 feature vectors by computing the sum of the squared residuals,

$$H(w, k) = \sum_{i=1601}^{2000} (y(i+1) - \tilde{m}(z(i);\ w,\ k))^2. \tag{4.3}$$

With $H(w, k)$ as objective function a 'best' weight a for each k was found by a cyclic coordinate search (Luenberger (1973, § 7.8)) over a region of positive quintuples, and finally a 'best' k was chosen by searching over the integers $k = 3$ to 6, using the best w for each k.

The ARMAX model, by definition, has the form

$$y(n+1) = \sum_{j=0}^{p} y(n-j)a(j) + \sum_{k=0}^{q} x(n-k)b(k) + \sum_{i=0}^{r} e(n-i)c(i). \tag{4.4}$$

The interpretation we give is that as before, $y(i)$ and $x(i)$s are the (observable) runoff and rainfall measurements. The $e(i)$ sequence is presumed to be unobservable white noise. For any given orders (p, q, r), we employed the Ljung and Soderstrom (1983) recursive maximum likelihood estimation algorithm to find the best parameters. We used our split sample, split as described above, to choose

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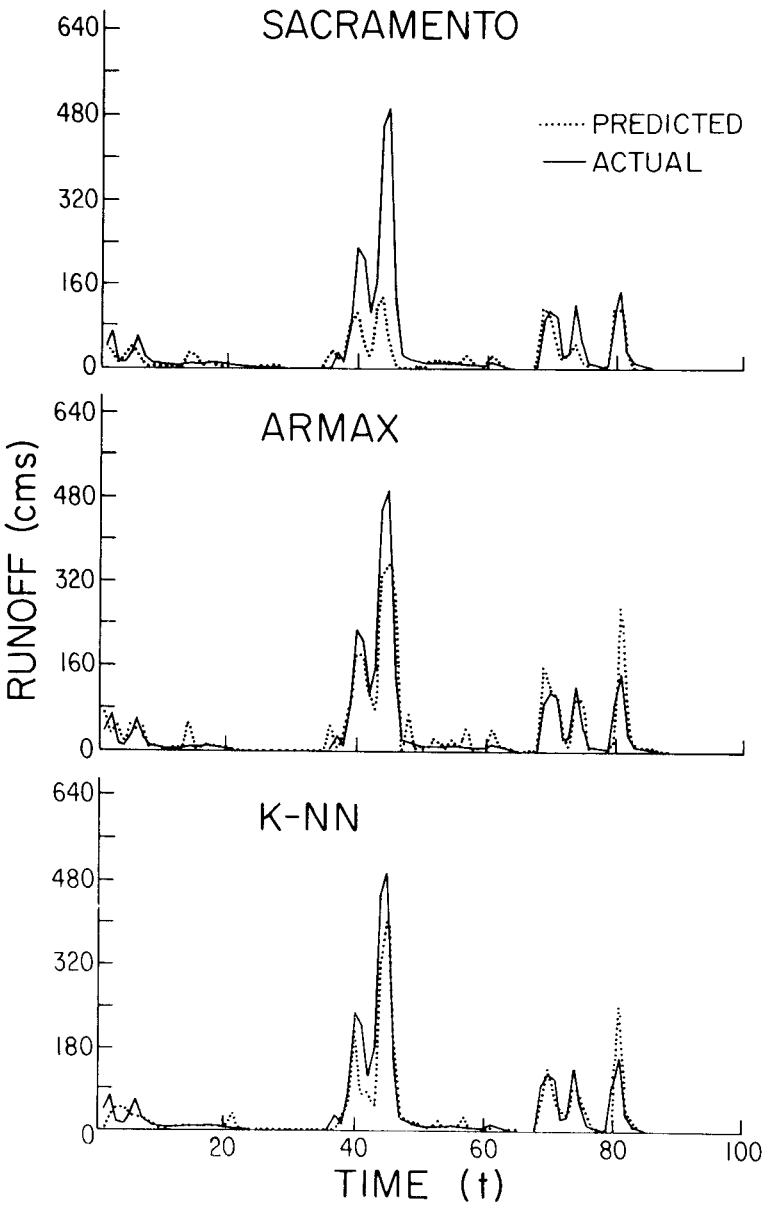


FIGURE 2. One day ahead prediction Bird Creek, days 2000–2100.

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TABLE I

SUM OF RESIDUAL SQUARE ERRORS OVER THE PREDICTION TEST SET

Sequence No.	k-NN	Sacramento	ARMAX
1–100	119,381	390,683	120,373
101–200	274,826	902,684	390,650
201–300	36,406	88,904	32,822

the 'best' (p, q, r) by calibrating on the first part, and measuring prediction error on the second. In this way, we concluded that the best ARMAX orders for our Bird Creek data were $(p, q, r) = (3, 3, 3)$.

Now finally, we compared the performance on the 300 point prediction set. The one day ahead prediction *vrs* flow graphs are shown for the first 100 days of the calibration set of Figure 2. The sum of the residual square errors over the complete prediction data set are presented in Table I.

One sees that the NN and ARMAX performances are comparable. But both are considerably better than the performance of the Sacramento model. Our conclusion, based on these and related findings, is that the NN method is not significantly worse than the standard second order methods on second-order theory home base: least squares problems. These standard methods are not applicable in more general decision problem frameworks, whereas in the manner of kernel regression (Yakowitz 1985), the NN method is.

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